Lecture 13
Challenges in Statistical Inference

Algorithms, AI & Data Science II

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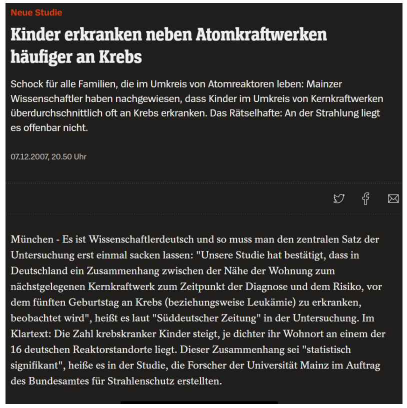
Motivation

- we introduced methods to **test statistical propositions** in empirical data
- we formulate **null (H_0) and alternative hypothesis (H_1)**
- distribution of **test statistic** under H_0 allows to assess **confidence in null hypothesis compared to alternative**
- we often use significance threshold α to **dichotomize confidence in null hypothesis** (i.e. we reject H_0 or not)
- use of p-values to **dichotomize decisions or discoveries** introduces many challenges
- introduces opportunities to **misuse and politizise data** that threaten credibility of science as a whole


Neue Studie

Kinder erkranken neben Atomkraftwerken häufiger an Krebs

Schock für alle Familien, die im Umkreis von Atomreaktoren leben: Mainzer Wissenschaftler haben nachgewiesen, dass Kinder im Umkreis von Kernkraftwerken überdurchschnittlich oft an Krebs erkranken. Das Rätselhafte: An der Strahlung liegt es offenbar nicht.

07.12.2007, 20.50 Uhr

München - Es ist Wissenschaftlerdeutsch und so muss man den zentralen Satz der Untersuchung erst einmal sacken lassen: "Unsere Studie hat bestätigt, dass in Deutschland ein Zusammenhang zwischen der Nähe der Wohnung zum nächstgelegenen Kernkraftwerk zum Zeitpunkt der Diagnose und dem Risiko, vor dem fünften Geburtstag an Krebs (beziehungsweise Leukämie) zu erkranken, beobachtet wird", heißt es laut "Süddeutscher Zeitung" in der Untersuchung. Im Klartext: Die Zahl krebskranker Kinder steigt, je dichter ihr Wohnort an einem der 16 deutschen Reaktorstandorte liegt. Dieser Zusammenhang sei "statistisch signifikant", heiße es in der Studie, die Forscher der Universität Mainz im Auftrag des Bundesamtes für Strahlenschutz erstellten.

image credit: www.spiegel.de, fair use

Notes:

- **Lecture 13: Challenges in Statistical Inference** 23.06.2025
- **Educational objective:** We highlight fallacies in the interpretation of p-values and introduce methods to avoid them. We discuss the difference between correlation and causation and show how we can use causal diagrams to avoid wrong interpretations of data.
  - p-values and Multiple Hypothesis testing
  - Basics of Causal Inference
  - Simpson's Paradox
- **Exercise sheet 07:** Hypothesis Testing and Causal Diagrams due 30.06.2025
- **Handout version:** *2025/06/23 10:00:29*

Notes:

The problem with p-values ...

- p-value $\neq$ prob. that null hypothesis is true
- p-value $=$ prob. to observe test statistic as extreme as in our data if H_0 was true
  - depends on **test statistic**
  - depends on **alternative hypothesis**
  - depends on **sample size and variance**
- **large sample size fallacy**
  - in small data, large effects can be insignificant
  - in large data, negligible effect can be highly significant
  - **significance $\neq$ relevance**
- responsible data scientists additionally consider **potential "damage"** of type I and type II errors


EDITORIAL

The ASA's Statement on p-Values: Context, Process, and Purpose

→ Ronald L. Wasserstein: The ASA Statement on p-Values: Context, Process, and Purpose, 2016

The salmon of doubt

- **functional MRI study** of salmon that was asked to judge pictures of human emotions
- study reveals **significant relationship** between human emotion and activity in specific brain area

problems

1. salmon?!?
2. salmon was dead

- what is wrong with this study?

> "The salmon was shown a series of photographs depicting human individuals in social situations with a specified emotional valence, either socially inclusive or socially exclusive. The salmon was asked to determine which emotion the individual in the photo must have been experiencing." → C. M. Bennett et al., Journal of Serendipitous and Unexpected Results, 2010

image credit: Ryan McGuire from Pixabay

Notes:

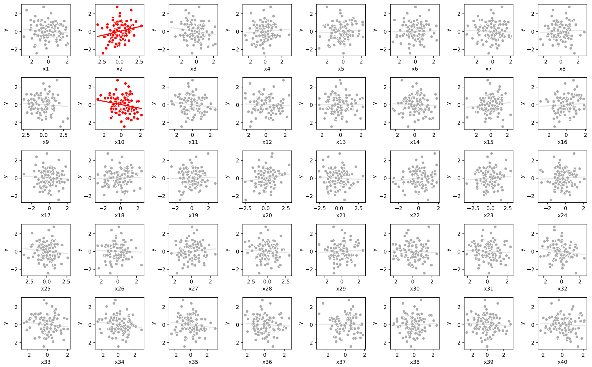
- Now that we have seen different examples for tests that are based on p-values, we conclude with some critical remarks regarding their interpretation.
- First, we repeat that the p-value is not the probability that the null hypothesis is true. It is rather a value that is based on the alternative hypothesis, the test statistic used, as well as the size and variance of our data set.
- Also, a small p-value means that the result is highly significant, but not that the result is relevant. We can have small data, where large effects are not significant, and we can have massive data sets where even the tiniest of effects can be highly significant. Clearly, large effects – even though we might not yet be sure about their significance – may be more relevant than significant but negligible effects! This is called the large sample size fallacy. Unfortunately, the academic publication landscape tends to be largely biased towards significant results, often neglecting effect size.
- When you make (or recommend) decisions based on p-values and significance thresholds, you should not only keep in mind the probability to make type I/II errors, but also what would be the implications of such errors! If the effect of wrongly rejecting a null hypothesis is catastrophic, you should favour very small p-values. But what if the effect of a type II error is much more severe than that of making a type I error?

Notes:

- Moreover, there are other issues in the interpretation of p-values that you must keep in mind, and that should make you skeptical.
- The example above introduces the famous "dead salmon study", which was published some years ago. This study was not meant seriously, it rather highlighting fallacies of hypothesis testing in high-dimensional data – and casts a critical light on data analytics techniques used in many functional MRI studies. It won the IgNobel prize 2012 and you can find it here:
  Bennett et al. "Neural Correlates of Interspecies Perspective Taking in the Post-Mortem Atlantic Salmon: An Argument For Proper Multiple Comparisons Correction" Journal of Serendipitous and Unexpected Results, 2010
- The study put a dead salmon in an MRI scanner, measuring the activity level of different brain regions, while it was shown pictures of humans displaying different emotional expressions. The study found a significant relationship between the type of emotion shown on the picture and the activity in one area of its (dead) brain. The study demonstrated how hypothesis tests can be misused to discover spurious patterns in the noise of fMRI data.

More data = more insights?

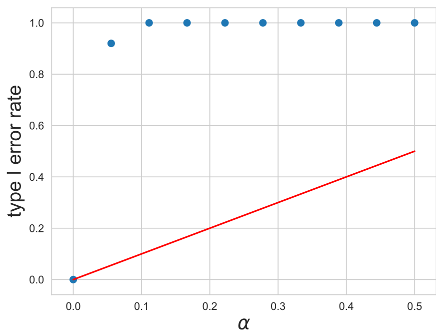
consider multi-variate data (X_i, Y) with **40 variables** X_i and variable Y, where X_i and Y are independent



t-test reveals that 2/40 linear relationships between (X_i, Y) are **statistically significant** for significance threshold $\alpha = 0.05$

Multiple hypothesis testing

- significance threshold α controls **type I error rate**, i.e. fraction of wrongly rejected null hypotheses
- wrongly rejected null hypothesis = **false discovery**
- for m independent tests, **expected number of false discoveries** is $m \cdot \alpha$
- for $m = 40$ tests with $\alpha = 0.05$ we expect two wrong rejections of H_0
- for **any significance threshold** α "discovery" is guaranteed for sufficiently large m



type I error rate for $m = 40$ t-tests of linear relationship

Notes:

- The example above illustrates the underlying problem in the context of linear regression. Here I have generated samples of 40 random variables X_i, as well as a single variable Y. There is no relationship whatsoever between any of them, i.e. all of them are independent. I then applied 40 separate linear regressions, using different X_i as independent and Y as dependent variable. We finally applied Student's t-test and report those independent variables X_i for which we find a significant linear relationship with Y for a significance threshold of $\alpha = 0.05$.
- The result should not surprise you, because we know that the type I error rate of each hypothesis test is 5 %. That is, if we repeat this test 40 times, we can expect to wrongly reject the null hypothesis that there is no relationship for two of the 40 null hypotheses, i.e. we expect that we make two false discoveries in our data set. This is an example where applying statistical methods to more data does not lead to more but rather to fewer insights, at least if we do not correct for the added probability to make wrong statements if we test hypotheses for more variables.

Notes:

- The example from the previous slide is a simple particularly simple case of the multiple hypothesis testing fallacy, which has become a major problem in data science. It is due to the fact that the **significance threshold α controls the type I error rate for a single test**, which means that the type I error rate is $m \cdot \alpha$ if we perform m independent tests.
- Clearly, no matter how small we choose the threshold α, we can always increase the number of tests m such that we are guaranteed to make a discovery, i.e. we reject at least one of the m null hypotheses.
- This is shown in the plot above, which shows the actual type I error rate in $m = 40$ tests of a linear relationship (in data where no such relationship exist) for different values of α. The red line shows

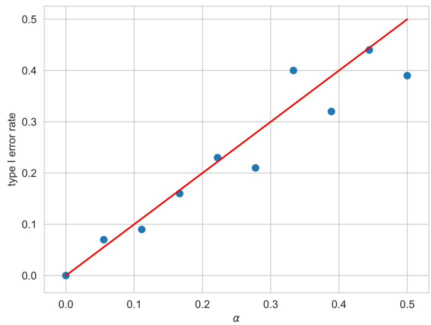
Bonferroni correction

- to maintain **expected type-I error rate** α for m tests, we must adjust significance threshold

$$\alpha' := \frac{\alpha}{m}$$

→ CE Bonferroni, 1935

- we treat "family" of m tests as a single test, i.e. we correct **family-wise error rate (FWER)**
- problem: fixing type-I error rate for m tests decreases **test power** for each individual test, i.e. we increase type-II error
- Bonferroni correction is **conservative approach** to correct for multiple tests

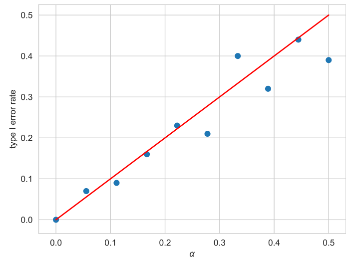


type I error rate for $m = 40$ t-tests of linear relationship with Bonferroni correction

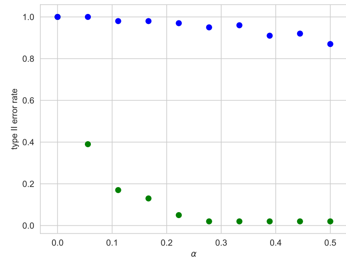
Notes:

- There is a simple method to address the multiple hypothesis testing problem. If we want to keep the interpretation of the significance threshold in terms of the type I error rate for m independent tests, we must fix the type I error rate $m \cdot \alpha$, i.e. we should decrease the threshold α as the number of tests grows. This is the idea behind the Bonferroni correction, which is the simplest way to avoid the multiple hypothesis testing fallacy.
- You see that in the dead salmon study, one should correct the significance threshold for the number of brain regions (i.e. the resolution of the fMRI scanner in terms of voxels), i.e. the number of null hypotheses that we test. Without this correction, we are virtually guaranteed to make a false discovery if we have enough voxel pairs for which we can repeat the test.
- But this simple solution comes at a hefty price: Fixing the type I error for m tests means that we treat m tests as a single test, which means that the type I error for each particular hypothesis is effectively decreased. This considerably increases the type II error rate, i.e. the power of the test decreases. In other words, we prevent false discoveries but we also prevent discoveries in general!

Type-I vs. type-II error



type I error rate for $m = 40$ t-tests of linear relationship with Bonferroni correction



type II error rate for $m = 40$ t-tests of linear relationship with (blue) and without (green) Bonferroni correction

false discovery rate

For m hypothesis tests, let r be the total number of rejections of H_0 and f the number of false rejections of H_0 (type I errors). The **false discovery rate** FDR is defined as

$$FDR := \frac{f}{r}$$

Notes:

- The experiment above illustrates this issue. The left plot shows the type I error rate for 40 hypothesis tests with a Bonferroni correction. The right plot shows the type II error rate for different values of α with (blue) and without (green) the Bonferroni correction in data where a linear relationship exists. The Bonferroni correction prevents us from detecting the relationship!
- The problem is that we cannot decrease the type II error without changing the interpretation of the significance threshold in terms of the type I error.
- A solution it to replace the type I error rate by the false discovery rate. It is defined as the fraction of false discoveries (i.e. wrongly rejected null hypotheses/type I errors) among the **total number of discoveries** (i.e. all rejected null hypotheses) in n tests. The difference to the type I and II error rate is subtle but important: If we only reject few hypotheses in a large number of tests, but most of those few rejections are type I errors, we have a small type I error rate, but a large false discovery rate (i.e. few of our tests are wrong, but most of those few discoveries are wrong). In other words, we fix the issue that the type I error rate does not account for the number of hypotheses that we reject, i.e. we can get a small type I error rate simply by not rejecting any hypotheses!

Controlling false discovery rate

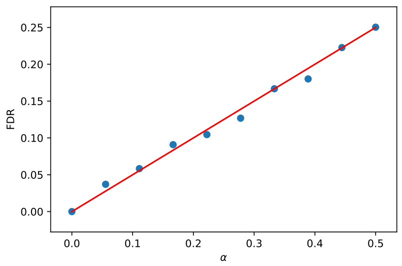
- we can adjust significance threshold α such that we control **false discovery rate** → Y Benjamini, Y Hochberg, 1995
- let m_f be the number of null hypotheses that should be rejected in m tests

Benjamini-Hochberg correction

1. let $p_1 \leq p_2 \ldots \leq p_m$ be the p-values of m tests of null hypotheses $H_1, \ldots, H_m$ against their corresponding alternatives
2. for given α find largest k such that $p_i \leq \frac{i}{m}\alpha$ for $i \leq k$
3. reject $H_1, \ldots, H_k$

- this correction yields **false discovery rate** of

$$FDR \leq \frac{m_f}{m}\alpha \leq \alpha$$



false discovery rate for $m = 20$ t-tests with **Benjamini-Hochberg correction** and $m_f = 10$

HARKing and p-hacking

- to HARK = **H**ypothesise **A**fter **R**esults are **K**nown
- often goes together with **data dredging** or **p-hacking**
- amplified by **publication and confirmation bias**
- further amplified by **clickbait media** and **selective reporting**

> "Subjects of the **chocolate intervention group** experienced the easiest and most successful weight loss. Even though the measurable effect of this diet occurred with a delay, the weight reduction of this group exceeded the results of the low-carb group by 10% after only three weeks (p = 0.04)."

how to avoid data misuse

- correct for multiple hypotheses
- report and interpret effect size
- theory- vs. data-driven research
- publication of registered reports

Notes:

- Using the false discovery rate, we can now introduce the Benjamini-Hochberg correction, which was introduced by Yoav Benjamini and Yosef Hochberg in 1995. Different from the Bonferroni correction it **changes the threshold α to reject a null hypothesis such that the false discovery rate in m tests is bounded above by α**. For this, we first compute the m p-values of our m tests and sort them in ascending order. We then determine the largest value k such that the first k p-values p_i are smaller than $\frac{i}{m}\alpha$ and reject the first k null hypotheses, while not rejecting the other null hypotheses.
- This correction yields a false discovery rate that is bounded above by α. The upper bound depends on the number of null hypotheses m_f that are false. If all of them are false ($m_f = m$) the upper bound is α, if none of them is false ($m_f = 0$) the upper bound is zero.
- The plot on the right shows the false discovery rate in $m = 20$ t-tests of linear relationships with $m_f = 10$ for different values of α, using the Benjamini-Hochberg correction. The red line marks the upper bound of $\frac{1}{2}\alpha$. We find that, due to the underlying conservative approach, the Bonferroni correction yields a small false discovery rate even for large values of α. Using the Benjamini-Hochberg correction, the threshold α instead controls the false discovery rate, which increases the test power for individual hypotheses.

Notes:

- HARKIng, data dredging and p-hacking, have become major challenges in data science. A problem with data-driven, exploratory studies is that we can (inadvertently) test multiple hypotheses and then only report (false) discoveries. While the multiple hypothesis testing fallacy still applies, this may not be obvious from the reported results. A related problem is p-hacking or data dredging, which refers to the practice of selectively analysing data such that we obtain the finding that we wanted to find. This makes it difficult to interpret empirical studies, and the problem is aggravated by publication pressure in science, the bias of journals to only accept positive results (i.e. selectively publishing discoveries rather than studies that failed to reject a null hypothesis), as well as clickbait journalism, selective reporting, and an often unhealthy urge of scientists to appear in media for the sake of publicity and career promotion.
- The mechanism behind this were demonstrated in a fake study on the allegedly positive health effects of chocolate. Despite making obvious mistakes in the data analysis (like multiple hypotheses testing without correcting the significance threshold), the study was readily published by a predatory journal and the wrong conclusions were happily picked up by major media outlets.
- Several practices to prevent such issues exist: Apart from using state-of-the-art data analytics techniques, we should honestly report and interpret effect size and not only significance. Doing research that is driven by theory and hypotheses rather than available data is another recipe against HARKing, since we fix the hypotheses that we test before collecting or analysing the data. To enforce this some journals are moving to registered reports, which requires researchers to submit research questions and hypotheses (i.e. study design) before collecting and analysing the data.

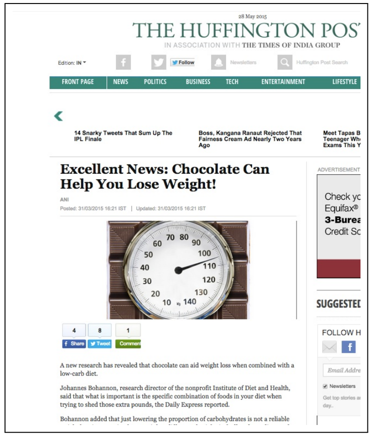
Decisions in multivariate problems

- so far, we only considered the question whether a **relationship** exists or not
- we are usually interested in relationships because we want to **influence outcomes**
- example: influence dependent variable Y by **manipulating independent variable** X

example

- X = chocolate consumption
- Y = weight

- does **correlation** between X and Y imply that **influencing X will change Y**?


28 May 2015

THE HUFFINGTON POS

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14 Snarky Tweets That Sum Up The IPL Finale

Boss, Kangana Ranaut Rejected That Fairness Cream Ad Nearly Two Years Ago

Meet Tapas B Teenager Wh Exams This Y

Excellent News: Chocolate Can Help You Lose Weight!

ANI

Posted: 31/03/2015 16:21 IST | Updated: 31/03/2015 16:21 IST

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A new research has revealed that chocolate can aid weight loss when combined with a low-carb diet.

Johannes Bohannon, research director of the nonprofit Institute of Diet and Health, said that what is important is the specific combination of foods in your diet when trying to shed those extra pounds, the Daily Express reported.

Bohannon added that just lowering the proportion of carbohydrates is not a reliable

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Recap: Covariance and Correlation

covariance of random variables

Let X_1 and X_2 be two random variables. The covariance $\text{Cov}(X_1, X_2)$ of X_1 and X_2 is defined as

$$\text{Cov}(X_1, X_2) := E\left[(X_1 - E(X_1))\,(X_2 - E(X_2))\right]$$

we have

- $\text{Cov}(X, X) = E\left[(X - E(X))^2\right] = \sigma^2 = \text{Var}(X)$ and
- $\text{Cov}(X_1, X_2) = \text{Cov}(X_2, X_1)$

- $\text{Cov}(X_1, X_2)$ captures how much X_1 and X_2 “vary together”
- we define symmetric **Pearson correlation coefficient**

$$\rho(X_1, X_2) := \frac{\text{Cov}(X_1, X_2)}{\sqrt{Var(X_1)}\sqrt{Var(X_2)}}$$

- Pearson correlation is **symmetric**, i.e. $\rho(X, Y) = \rho(Y, X)$

Notes:

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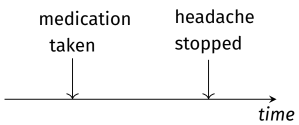
Example: effect of medical treatment

- consider (fictional) data from study on the effects of a medical treatment in **patients suffering from headache**
- each patient records ...
  1. whether they took medicine (and if yes when), and
  2. whether they suffered from a headache one hour later
- results of study are given in following table

	med. (1)	no med. (0)	total
stop (1)	54	10	64
no stop (0)	5	23	28
total	59	33	92

- is there a correlation between taking medication and the fact that the headache stopped?

image credit: Bing image creator



Group exercise 13-01

- Consider the data from the previous slide. Compute **covariance** and **Pearson correlation coefficient** of the binary random variables X (took medication) and Y (headache stopped).
- Would you say that taking medication and the fact that the headache stopped are correlated?

$$E(X) = \frac{33}{92} \cdot 0 + \frac{59}{92} \cdot 1 = \frac{59}{92}$$
$$E(Y) = \frac{28}{92} \cdot 0 + \frac{64}{92} \cdot 1 = \frac{64}{92}$$
$$\mathrm{Cov}(X,Y) = E\left[(Y - E(Y))(X - E(X))\right]$$
$$\mathrm{Cov}(X,Y) = E\left[\left(Y - \frac{64}{92}\right)\left(X - \frac{59}{92}\right)\right]$$
$$\mathrm{Cov}(X,Y) = \frac{54}{92}\left(1 - \frac{64}{92}\right)\left(1 - \frac{59}{92}\right) + \frac{10}{92}\left(1 - \frac{64}{92}\right)\left(0 - \frac{59}{92}\right) + \frac{5}{92}\left(0 - \frac{64}{92}\right)\left(1 - \frac{59}{92}\right) + \frac{23}{92}\left(0 - \frac{64}{92}\right)\left(0 - \frac{59}{92}\right) \approx 0.14$$

$$Var(X) = E\left[(X - E(X))^2\right] = \frac{59}{92}\left(1 - \frac{59}{92}\right)^2 + \frac{33}{92}\left(0 - \frac{59}{92}\right)^2$$
$$Var(Y) = E\left[(Y - E(Y))^2\right] = \frac{64}{92}\left(1 - \frac{64}{92}\right)^2 + \frac{28}{92}\left(0 - \frac{64}{92}\right)^2$$
$$\rho = \frac{\mathrm{Cov}(X,Y)}{\sqrt{Var(X)}\sqrt{Var(Y)}} \approx \frac{0.14}{\sqrt{0.23} \cdot \sqrt{0.21}} \approx 0.63 > 0$$

The positive correlation coefficient indicates that X and Y are positively correlated.

Notes:

Notes:

Correlation vs. Causation

- decision-making requires understanding of **cause-effect relationships**
- but: correlation $\rho(X, Y) > 0$ does **not** imply **causal relationship** from X to Y
- to reliably study cause and effect, in principle we would require a **counterfactual analysis**
  - intervention: study outcome if we change variable in the system, e.g. we administer medication to patient
  - counterfactual: study outcome that would have occurred if we had not made the intervention, i.e. we did not administer the medication
- in practice, we often address this using **controlled trials**, where **control group** is used as proxy for counterfactual

example

- Group 1 was **at home**, had access to medicine, and took a nap after taking the medicine.
- Group 2 was **at work** and could neither take the medicine nor take a nap.

DALL-E image depicting the "multiverse"

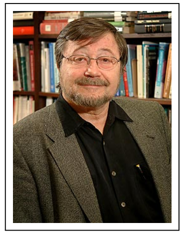
image credit: Bing image creator

Causal diagrams

- we can model causal relationships with **causal diagrams**, i.e. **directed graphs** where nodes represent variables
- dashed (undirected) edges indicate **correlation**
- directed edges indicate (possible) **causal relationship**
- causal relationships form a **directed acyclic graph**
- **graphical modelling framework** that allows to integrate statistical information and **domain-specific knowledge**

  → J Pearl, 1995
- basis for more advanced **causal inference** techniques

  → advanced courses



Judea Pearl, born 1936

Turing Award 2011

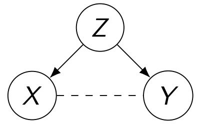


image credit: American Academy of Arts and Sciences, fair use

Notes:

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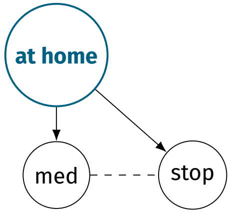
Confounders

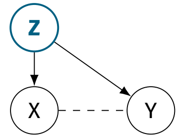
- consider **causal diagram** with random variables indicating whether patients took medication (`med`) and whether headache stopped (`stop`)
- dashed (undirected) edge indicates **correlation between variables** `med` and `stop`
- independent and dependent variable can both be causally influenced by (unobserved) **confounding factor**
- can lead to **spurious correlation** even if independent and dependent variable are **not causally related**
- confounder can invalidate **aggregate analysis** of data
- we need to analyse data **for each value of confounder separately** → later more



example

- patients who are at home **can take medication**
- patients who are at home **can also take a nap**, which reduces the headache
- **being at home** is a confounding factor

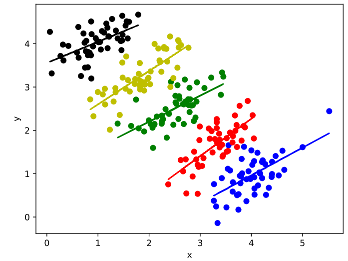
Controlling for confounders



- to isolate relationship between two variables X and Y, we must **control for possible confounders** Z
- for confounders with **categorical values** we **stratify data**, i.e. we model relationships between X and Y separately for groups of data where Z is constant
- for confounders with **continuous values** we apply **multivariate analysis**, e.g. multiple linear regression

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \epsilon$$

- fitted slope β_1 captures relationship between X and Y, while **accounting for relationship between Z and Y**
- controlling for confounder can **reverse inferred relationship** → Simpson's paradox



a stratified linear regression, where we control for the groups of data (indicated by colors), yields positive linear relationships while an aggregate analysis of the same data would yield a strong negative relationship

image credit: Luka Petrović

Notes:

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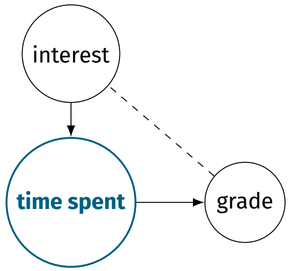
Mediators

- consider a (fictional) study on student grades and interest in a subject

example: exam grades

- we find positive correlation between a **student's interest** in a subject at the start of the semester and the **grade** in the final exam

- is it really the **interest** that directly causally influences the grade in the final exam?
- independent variable X can causally influence variable Z, which causally influences dependent variable Y
- we call Z a **mediator** between X and Y



Collider

- consider a (fictional) study on acting skills, attractiveness, and success of actors

example

- we assume that **attractiveness** and **acting skills** are not related
- but both causally influence success

- if two variables X and Y causally influence a third variable Z, we call Z a **collider** for X and Y
- controlling for a collider Z introduces **spurious correlation** between X and Y
- example: **analysis of successful actors** would indicate spurious (negative) correlation between attractiveness and acting skills

DALL-E generated image of successful Hollywood actor that is not attractive :-)

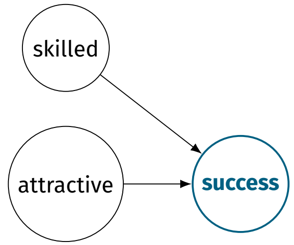


image credit: Bing image creator

Notes:

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Application: Simpson's paradox

- during a pandemic, health officials must **decide which of two medications they should buy**
- they are shown data from a **clinical study** that investigated the fatality rate among patients, depending on the patients' condition at the start of the treatment

	patient condition		
medication	mild	severe	aggregate
A	20/1100 (1.8%)	2400/9830 (24.4%)	2420/10930 (22.1%)
B	303/10400 (2.9%)	289/850 (34%)	592/11250 (5.3%)

- there is only sufficient resources for **one of the two medications**, so which of the two would you choose and why?
- instance of **Simpson's paradox**, where effect is reversed if we aggregate data on multiple groups

Scenario 1: Mediator

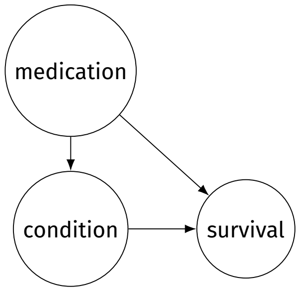
- decision depends on **domain knowledge** that we can represent as **causal diagram**

example knowledge

- medication B can be administred quickly, while medication A requires an additional test
- thus, **patients have to wait longer** and are more likely to be in a severe condition once they are finally administered medication A

- patient condition is **mediator** between medication and survival
- to study full causal effect of medication on survival, we should **not control** for condition
- **choose medication B** with **lowest aggregate fatality rate**

	patient condition		
medication	~~mild~~	~~severe~~	aggregate
A	~~20/1100 (1.8%)~~	~~2400/9830 (24.4%)~~	2420/10930 (22.1%)
B	~~303/10400 (2.9%)~~	~~289/850 (34%)~~	592/11250 (5.3%)



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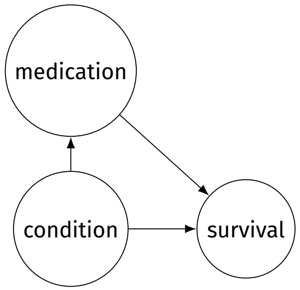
Scenario 2: Confounder

- decision depends on **domain knowledge** that we can represent as **causal diagram**

example knowledge

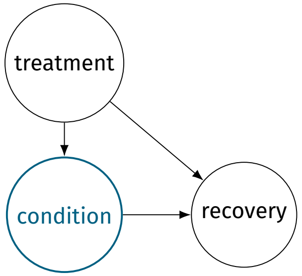
- medical personel assigns medication to patients based on their condition
- medication A is **selectively administered to patients in severe condition**

	patient condition		
medication	mild	severe	aggregate
A	20/1100 (1.8%)	2400/9830 (24.4%)	2420/10930 (22.1%)
B	303/10400 (2.9%)	289/850 (34%)	592/11250 (5.3%)

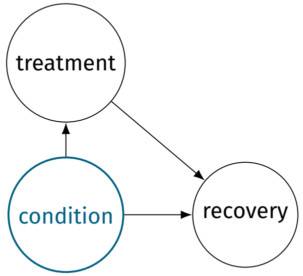


- patient condition is a **confounder** that we must control for
- controlling for confounder, we should **choose medication A** that has lowest fatality rate in each patient group

To control, or not to control ...



mediator scenario



confounder scenario

decision

- we do not control for patient condition
- choose B, since it has lower aggregate fatality rate

explanation

lower fatality rates of A for different groups can be explained by the fact that it is a better treatment once it is finally administered

decision

- we control for patient condition
- choose A, as it has lower fatality rate for both patient groups

explanation

lower aggregate fatality rate of B can be explained by the fact that it is assigned more often to patients in mild condition

Notes:

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In summary

- we learned how to **use data to test statistical propositions**
- **problematic fixation** on p-values in empirical sciences and **problematic public understanding of significance**
- apart from opportunities, big data introduces challenges that threaten **internal and external validity** of findings
- decision-making requires domain knowledge on **possible causal relationships** (e.g. using causal diagrams)

> HARKing + lack of causal reasoning + publication bias + "publish or perish" + selective reporting by politicized outlets + clickbait media = **existential threat for credibility of science**

- **be critical** whenever you read "a new study found a significant correlation …"


Neue Studie

Kinder erkranken neben Atomkraftwerken häufiger an Krebs

Schock für alle Familien, die im Umkreis von Atomreaktoren leben: Mainzer Wissenschaftler haben nachgewiesen, dass Kinder im Umkreis von Kernkraftwerken überdurchschnittlich oft an Krebs erkranken. Das Rätselhafte: An der Strahlung liegt es offenbar nicht.

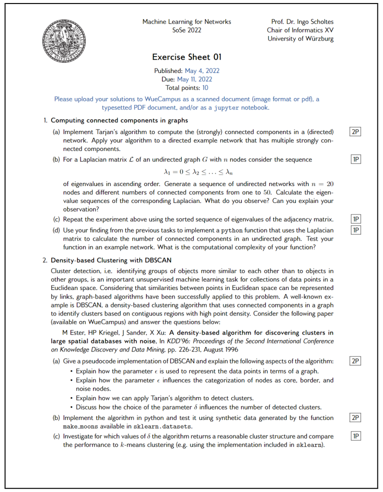
07.12.2007, 20.50 Uhr

München - Es ist Wissenschaftlerdeutsch und so muss man den zentralen Satz der Untersuchung erst einmal sacken lassen: "Unsere Studie hat bestätigt, dass in Deutschland ein Zusammenhang zwischen der Nähe der Wohnung zum nächstgelegenen Kernkraftwerk zum Zeitpunkt der Diagnose und dem Risiko, vor dem fünften Geburtstag an Krebs (beziehungsweise Leukämie) zu erkranken, beobachtet wird", heißt es laut "Süddeutscher Zeitung" in der Untersuchung. Im Klartext: Die Zahl krebskranker Kinder steigt, je dichter ihr Wohnort an einem der 16 deutschen Reaktorstandorte liegt. Dieser Zusammenhang sei "statistisch signifikant", heiße es in der Studie, die Forscher der Universität Mainz im Auftrag des Bundesamtes für Strahlenschutz erstellten.

image credit: www.spiegel.de, fair use

Exercise sheet 07

- **seventh exercise sheet** will be released today
- solutions are due **June 30th** (via WueCampus)
- present your solution to earn bonus points


Machine Learning for Networks
SoSe 2022

Prof. Dr. Ingo Scholtes
Chair of Informatics XV
University of Würzburg

Exercise Sheet 01

Published: May 4, 2022
Due: May 11, 2022
Total points: 10

Please upload your solutions to WueCampus as a scanned document (image format or pdf), a typesetted PDF document, and/or as a `jupyter` notebook.

1. **Computing connected components in graphs**
   (a) Implement Tarjan's algorithm to compute the (strongly) connected components in a (directed) network. Apply your algorithm to a directed example network that has multiple strongly connected components. 2P
   (b) For a Laplacian matrix $\mathcal{L}$ of an undirected graph G with n nodes consider the sequence 1P
   $$\lambda_1 = 0 \leq \lambda_2 \leq \ldots \leq \lambda_n$$
   of eigenvalues in ascending order. Generate a sequence of undirected networks with $n = 20$ nodes and different numbers of connected components from one to 50. Calculate the eigenvalue sequences of the corresponding Laplacian. What do you observe? Can you explain your observation?
   (c) Repeat the experiment above using the sorted sequence of eigenvalues of the adjacency matrix. 1P
   (d) Use your finding from the previous tasks to implement a `python` function that uses the Laplacian matrix to calculate the number of connected components in an undirected graph. Test your function in an example network. What is the computational complexity of your function? 1P
2. **Density-based Clustering with DBSCAN**
   Cluster detection, i.e. identifying groups of objects more similar to each other than to objects in other groups, is an important unsupervised machine learning task for collections of data points in a Euclidean space. Considering that similarities between points in Euclidean space can be represented by links, graph-based algorithms have been successfully applied to this problem. A well-known example is DBSCAN, a density-based clustering algorithm that uses connected components in a graph to identify clusters based on contiguous regions with high point density. Consider the following paper (available on WueCampus) and answer the questions below:
   M Ester, HP Kriegel, J Sander, X Xu: **A density-based algorithm for discovering clusters in large spatial databases with noise**, In *KDD'96: Proceedings of the Second International Conference on Knowledge Discovery and Data Mining*, pp. 226-231, August 1996
   (a) Give a pseudocode implementation of DBSCAN and explain the following aspects of the algorithm: 2P
   - Explain how the parameter ϵ is used to represent the data points in terms of a graph.
   - Explain how the parameter ϵ influences the categorization of nodes as core, border, and noise nodes.
   - Explain how we can apply Tarjan's algorithm to detect clusters.
   - Discuss how the choice of the parameter δ influences the number of detected clusters.
   (b) Implement the algorithm in python and test it using synthetic data generated by the function `make_moons` available in `sklearn.datasets`. 2P
   (c) Investigate for which values of δ the algorithm returns a reasonable cluster structure and compare the performance to k-means clustering (e.g. using the implementation included in `sklearn`). 1P

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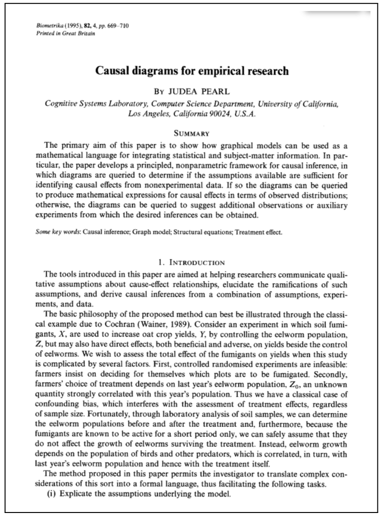
Self-study questions

1. Explain the problem that arise when testing multiple hypotheses.
2. What is the type-I error rate for a family of m hypothesis tests if we apply a significance threshold of α.
3. What is the Bonferroni correction? What type-I error rate does it have for a family of m tests?
4. How does the Bonferroni correction influence the type-II error rate of a family of m hypothesis tests?
5. What is the false-discovery rate in a family of m hypothesis tests?
6. What is the Benjamini-Hochberg correction? What is the false discovery rate if we use a significance threshold of α?
7. Explain HARKing and how can it be mitigated with registered reports?
8. Give an example for a confounding factor that leads to a spurious correlation.
9. Draw causal diagrams that illustrate a collider, confounder, and mediator variable.
10. Explain why we should not control for a mediator variable Z between X and Y when we want to study the (causal) relationship between X and Y.
11. What is Simpson's paradox? How can we use causal diagrams to address it?

References

reading list

- CE Bonferroni: **Il calcolo delle assicurazioni su gruppi di teste**, Studi in Onore del Professore Salvatore Ortu Carboni, 1935.
- JP Shaffer: **Multiple Hypothesis Testing**, Ann. Rev. Psych. 46, 1995
- TV Perneger: **What's Wrong with Bonferroni Adjustments**, Brit. Med. J. 316, 1998
- RG Miller: **Simultaneous Statistical Inference**, Springer, 1966
- Y Benjamini, Y Hochberg: **Controlling the false discovery rate: a practical and powerful approach to multiple testing**, Journal of the Royal Statistical Society, 1995
- J Pearl: **Causal Diagrams for Empirical Research**, Biometrika, 1995
- CM Bennet et al: **Neural Correlates of Interspecies Perspective Taking in the Post-Mortem Atlantic Salmon: An Argument For Proper Multiple Comparisons Correction**, Journal of Serendipitous and Unexpected Results, 2009
- Judea Pearl: **Causal Inference in Statistics: An Overview**, Statistics Surveys, Vol. 6, 2009
- J Pearl: **Causal Inference in Statistics - A Primer**, Wiley, 2016
- J Pearl, D MacKenzie: **The Book of Why: The New Science of Cause and Effect**, Basic Books, 2018
- P Ding: **A First Course in Causal Inference**, arXiv 2305.18793, 2023


Biometrika (1995), **82**, 4, pp. 669–710
Printed in Great Britain

Causal diagrams for empirical research

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SUMMARY

The primary aim of this paper is to show how graphical models can be used as a mathematical language for integrating statistical and subject-matter information. In particular, the paper develops a principled, nonparametric framework for causal inference, in which diagrams are queried to determine if the assumptions available are sufficient for identifying causal effects from nonexperimental data. If so the diagrams can be queried to produce mathematical expressions for causal effects in terms of observed distributions; otherwise, the diagrams can be queried to suggest additional observations or auxiliary experiments from which the desired inferences can be obtained.

Some key words: Causal inference; Graph model; Structural equations; Treatment effect.

1. INTRODUCTION

The tools introduced in this paper are aimed at helping researchers communicate qualitative assumptions about cause-effect relationships, elucidate the ramifications of such assumptions, and derive causal inferences from a combination of assumptions, experiments, and data.

The basic philosophy of the proposed method can best be illustrated through the classical example due to Cochran (Wainer, 1989). Consider an experiment in which soil fumigants, X, are used to increase oat crop yields, Y, by controlling the eelworm population, Z, but may also have direct effects, both beneficial and adverse, on yields beside the control of eelworms. We wish to assess the total effect of the fumigants on yields when this study is complicated by several factors. First, controlled randomised experiments are infeasible: farmers insist on deciding for themselves which plots are to be fumigated. Secondly, farmers' choice of treatment depends on last year's eelworm population, Z_0, an unknown quantity strongly correlated with this year's population. Thus we have a classical case of confounding bias, which interferes with the assessment of treatment effects, regardless of sample size. Fortunately, through laboratory analysis of soil samples, we can determine the eelworm populations before and after the treatment and, furthermore, because the fumigants are known to be active for a short period only, we can safely assume that they do not affect the growth of eelworms surviving the treatment. Instead, eelworm growth depends on the population of birds and other predators, which is correlated, in turn, with last year's eelworm population and hence with the treatment itself.

The method proposed in this paper permits the investigator to translate complex considerations of this sort into a formal language, thus facilitating the following tasks.

(i) Explicate the assumptions underlying the model.

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